

Artificial Intelligence (AI) represents one of the most transformative technology developments of the twenty-first century. By enabling computer systems to perform complex tasks that historically required human cognitive capabilities—such as visual perception, speech recognition, decision-making, and translation between languages—AI is reshaping global industries. Large language models, a subset of generative AI, utilize deep learning neural networks trained on vast datasets to synthesize text, write software code, and analyze structured data with remarkable speed and coherence. As these technologies continue to advance, standard methodologies in academic research, education, and software development are undergoing a fundamental paradigm shift.